

Problem Set 3.

1). $H_n(\mathbb{C}) \rightarrow$ set of $n \times n$
 Hermitian
 matrices.

i.e.,

$$H_n(\mathbb{C}) = \{ A \in \mathbb{C}^{n \times n} : A^H = A \}$$

$$A^H = (\overline{A})^T$$

$$\therefore A = \begin{bmatrix} a & x+iy \\ x-iy & b \end{bmatrix}$$

$$\therefore \overline{A} = \begin{bmatrix} a & x-iy \\ x+iy & b \end{bmatrix}$$

$$(\overline{A})^T = \begin{bmatrix} a & x+iy \\ x-iy & b \end{bmatrix}$$

$$= \underline{A}$$

* If $H_n(\mathbb{C})$ is a vector space over F , it is

- i) Closed under addition
- ii) Closed under scalar multiplication

~~i) $F = \mathbb{C}$ (set of complex no.s)~~

i) Closure under addition,

for $X, Y \in H_n(\mathbb{C})$,

$$\cancel{(X+Y)^H}$$

$$X = \begin{bmatrix} a_1 & c_1 + id_1 \\ c_1 - id_1 & b_1 \end{bmatrix}$$

$$Y = \begin{bmatrix} a_2 & c_2 + id_2 \\ c_2 - id_2 & b_2 \end{bmatrix}$$

$$\therefore X+Y = \begin{bmatrix} a_1+a_2 & (c_1+c_2) + i(d_1+d_2) \\ (c_1+c_2) - i(d_1+d_2) & b_1+b_2 \end{bmatrix}$$

$$(\overline{x+y}) = \begin{bmatrix} a_1 + ia_2 & (c_1 + ic_2) - i(d_1 + id_2) \\ (c_1 + ic_2) + i(d_1 + id_2) & b_1 + ib_2 \end{bmatrix}$$

$$(\overline{x+y})^T = \begin{bmatrix} a_1 + ia_2 & (c_1 + ic_2) + i(d_1 + id_2) \\ (c_1 + ic_2) - i(d_1 + id_2) & b_1 + ib_2 \end{bmatrix}$$

$$= x+y$$

$$\therefore \text{ as } (x+y)^H = x+y,$$

$$\therefore x+y \in H_n(c) \text{ for } x \in H_n(c) \text{ and } y \in H_n(c).$$

$\therefore$, its ~~closed~~ closed under addition.

ii). Closure under scalar multiplication

1. if $F = \mathbb{C}$ (complex no.s)

Let $\alpha = m + in$.

$$\therefore X = \begin{bmatrix} a & c + id \\ c - id & b \end{bmatrix}$$

$$\alpha X = \begin{bmatrix} (m + in)(a) & (c + id)(m + in) \\ (c - id)(m + in) & b(m + in) \end{bmatrix}$$

$$(\overline{\alpha X}) = \begin{bmatrix} (m - in)(a) & (c - id)(m - in) \\ (m - in)(c + id) & b(m - in) \end{bmatrix}$$

$$(\overline{\alpha X})^T = \begin{bmatrix} (m - in)(a) & (c + id)(m - in) \\ (c - id)(m + in) & b(m - in) \end{bmatrix}$$

$\neq \alpha X$.

$$\therefore (\overline{\alpha X})^T = (\overline{\alpha} \overline{X})^T = \overline{\alpha} (\overline{X})^T = \overline{\alpha} X$$

But $\alpha \neq \overline{\alpha}$ for $a \neq b$
 if $b \neq 0$.

$$\therefore \alpha X \notin H_n(\mathbb{C}).$$

So, it's NOT closed under scalar multiplication.

$\therefore H_n(\mathbb{C})$ is NOT a vector space over field $\mathbb{C}$.

2). $F = \mathbb{R}$ (real no.s).

~~$$\text{if } (\overline{\alpha X})^T = \alpha X$$~~

$$\begin{aligned} (\overline{\alpha X})^T &= (\overline{\alpha} \overline{X})^T \\ &= \overline{\alpha} (\overline{X})^T \\ &= \overline{\alpha} X. \end{aligned}$$

So, $\overline{\alpha} X = \alpha X$ for $\alpha \in \mathbb{R}$ if
 $H_n(\mathbb{C})$ is a
 vector space
 over $\mathbb{R}$.

for which,

$$\bar{\alpha} = \alpha.$$

This holds true for all real α as $\alpha \in \mathbb{R} = a + ib$, where $b = 0$.

$$\therefore a + ib = a - ib = a.$$

$$\therefore \alpha = \bar{\alpha}.$$

YES,

$\therefore H_n(\mathbb{C})$ is a vector space over field $\mathbb{R}$, as it's closed under scalar multiplication.

2). Given: Let W_1 and W_2 be ~~any~~ subspaces of vector space V , such that $W_1 \cup W_2$ is also a subspace.

To prove: Either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Proof:

Proof by contradiction:

Let ~~$W_1 \cap W_2 = \emptyset$~~ ,
 $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$,

$\therefore$ There exists some:

- i). $x \in W_1$, $x \notin W_2$, i.e. $x \in W_1 \setminus W_2$
- ii). $y \in W_2$, $y \notin W_1$, i.e. $y \in W_2 \setminus W_1$

Proof:

as $W_1 \cap W_2$ is a subspace,
and $x, y \in (W_1 \cap W_2)$

(as $x \in W_1$ and $y \in W_2$).

$\therefore$ By closure under $+$,

$$n+y \in (W, \cap W_2)$$

so, either $n+y \in W_1$,

or $n+y \in W_2$

or $n+y \in W_1$ and $n+y \in W_2$

$\S$:

a). $n+y \in W_1$.

$n \in W_1$ $\therefore$ due to inverse under $+$,

$$-n \in W_1$$

$$\therefore (n+y) + (-n) \in W_1,$$

$\S$ as $(n+y) \in W_1$ and $(-n) \in W_1$,

$\therefore$ by closure under addition $\S$

$$\therefore n+y-n \in W_1$$

$$0+y \in W_1$$

$$\boxed{y \in W_1}$$

But, this is in contradiction

with statement (ii).

b). $(n+y) \in W_2$

$$\begin{aligned} y &\in W_2 \\ (-y) &\in W_2 \end{aligned}$$

$$\therefore (n+y) + (-y) \in W_2$$

$$\therefore n \in W_2.$$

This is in contradiction with statement (i).

$\therefore$ By contradiction, the above assumption is FALSE.

$\therefore$ Either, there exists no n such that $n \in W_1, n \notin W_2$

or there exists no y ,

such that $y \in W_2, y \notin W_1$.

$\therefore$ Either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$ or both ($W_1 = W_2$)

Hence proved.

3) Given: $W = \cap_{\alpha} W_{\alpha}$, where
 $\{W_{\alpha}\}_{\alpha}$ are subspaces.

$\alpha, \beta \in W$ and c is
 a scalar.

To prove: $c\alpha + \beta \in W$.

Proof: W is intersection of all
 W_{α} .

So, for $\alpha, \beta \in W$,

α, β MUST also belong to
 EACH W_{α} in W ,

as they are in the intersecti-
 on of all W_{α} .

Hence, $\forall W_{\alpha}$ in W ,

$\alpha, \beta \in W_{\alpha}$.

Now, W_{α} is a subspace

$\therefore$ it's

Closed W

i). Closed under scalar multiplication

$\therefore$ for $\alpha \in W_\alpha$,

$$c\alpha \in W_\alpha.$$

ii). Closed under addition

$\therefore$ for $\alpha \in W_\alpha$ and $\beta \in W_\alpha$,

$$c\alpha + \beta \in W_\alpha.$$

for all W_α in W .

Now, as $c\alpha + \beta$ belongs to all W_α ,

they must belong in the intersection of these W_α .

$$\therefore c\alpha + \beta \in \bigcap W_\alpha$$

$$\boxed{c\alpha + \beta \in W}$$

Hence proved.

4) Given: F be a field and S be a set such that $S \neq \emptyset$

$$V = \{ f: S \rightarrow F \}$$

$$(f+g)(s) = f(s) + g(s)$$

$$(\cancel{f+g})(s) \rightarrow f$$

$$(cf)(s) = cf(s), \text{ where } c \text{ is a scalar.}$$

To prove: V is a vector space.

Proof:

For V to be a vector space, V must be:

1). Closed under $+$.

for $s \in S$, $f(s), g(s) \in F$

$$(f+g)(s) = f(s) + g(s) \in F$$

Now, V is all functions from $S \rightarrow F$

$$\therefore f(s) + g(s) = h(s) \in V.$$

$\therefore$ it's closed under $+$.

ii). Closure under scalar multiplication

for $c \in C$ (scalars),
 $f(s) \in V$

$$(cf)(s) = c(f(s)).$$

Now, $c(f(s)) = h'(s) \in V$

as $c(f(s)) : f \rightarrow s$.

$\therefore$ it's closed under scalar $\times$.

~~iii)~~

iii). Additive identity.

$$0(s) = 0_f \text{ for } \forall s \in S,$$

$$\therefore 0 \in V,$$

$\therefore$ for $f \in V, s \in S$

$$(f+0)s = f(s) + 0_f = f(s).$$

iv). Additive inverse.

~~Let $y \in V$~~

Let for $(-y) \in V$,

$$(-y)(s) = -(ys)$$

$\therefore$ scalar (-1) multiplied

$\therefore$ for $f \in V$.

$$\begin{aligned} (f + (-f))(s) &= f(s) + (-f)(s) \\ &= f(s) - f(s) \\ &= 0_f. \end{aligned}$$

v). Commutativity of $+$.

$$(f+g)s = f(s) + g(s)$$

$$(g+f)s = g(s) + f(s)$$

For functions $f(s)$ and $g(s) \in \mathbb{R}V$

$$f(s) + g(s) = g(s) + f(s)$$

$$\therefore (f+g)(s) = (g+f)(s).$$

vi) associativity of $+$.

$$\begin{aligned} ((f+g)+h)(s) &= (f+g)(s) + h(s) \\ &= f(s) + g(s) + h(s) \end{aligned}$$

$$\begin{aligned} (f+(g+h))(s) &= f(s) + (g+h)(s) \\ &= f(s) + g(s) + h(s) \\ &= ((f+g)+h)(s). \end{aligned}$$

$\therefore$ It's associative under $+$.

vii) $\times$ Identity.

Let for $c = 1$

$$(1f)(s) = 1(f(s)) = f(s)$$

viii) ~~for~~ For $\alpha, \gamma \in C$ (scalars)

$$\text{Prove: } (\alpha\gamma)(f(s)) = \alpha(\gamma f(s))$$

$$\begin{aligned} \text{Proof: } (\alpha\gamma)(f(s)) &= (\alpha\gamma f)(s) \\ &= \alpha(\gamma f)(s) \end{aligned}$$

(ix). Distributivity over ~~scalar~~ ^{vector} +.

$$\begin{aligned} c(f+g)(s) &= c(f(s) + g(s)) \\ &= cf(s) + cg(s) \\ &= (cf+cg)(s). \end{aligned}$$

x). Distributivity over scalar +.

~~$$(c+d)f(s) = c$$~~

$$\begin{aligned} (c+d)f(s) &= (c+d)f(s) \\ &= c(f(s)) + d(f(s)) \\ &= (cf+df)(s). \end{aligned}$$

Hence, as V satisfies all the above properties,

V is a vector space over scalars C .

where $V: f \rightarrow r$

$$V = \{ f: S \rightarrow Y \}$$

5) Given: S is set of n tuples,
 $(n_1, n_2; \dots, n_n)$

where $n_1 = 1 + n_2$ ($n \geq 2$)

Solⁿ:

For S to be a subpose,

$\exists 0_n \in S$ such that for $\forall x \in S$

$$x + 0 = x$$

$$\text{So, } n_1 \geq 0$$

$$n_2 \geq 0$$

$$\vdots$$

$$n_n \geq 0$$

$$\text{But } n_1 = 1 + n_2$$

$$\therefore \text{if } n_2 \geq 0$$

$$n_1 \geq 1.$$

$\therefore 0_n \notin S$. $\therefore$ It cannot have an additive identity
 $(0, 0, \dots, 0) \in S$

It's NOT a subspace.

Further, if $n_1 = 1 + n_2$

for $x, y \in S$

$$x = (1 + n_2, n_2, n_3, \dots, n_n)$$

$$y = (1 + y_2, y_2, y_3, \dots, y_n)$$

$$\therefore x + y = (2 + n_2 + y_2, n_2 + y_2, \dots, n_n + y_n)$$

$$\begin{aligned} \text{Now, } (x + y)_1 &= 2 + n_2 + y_2 \\ &= 2 + (n + y)_2 \\ &\neq 1 + (n + y)_2 \end{aligned}$$

$$\therefore (x + y) \notin S.$$

$\therefore$ closure under + also fails

Hence proved, NO, S is NOT a subspace.

6) Given: $W_1, \dots, W_k$ are subspaces of V .

$$W = W_1 + \dots + W_k$$

To prove: 1). W is a subspace of V .
2). W is the subspace spanned by $W_1 \cup W_2 \cup \dots \cup W_k$.

Proof:

1). for W to be a subspace of V ,

it MUST HAVE

i). Be non empty (have zero vector)

as $W_i, i \in [1, k]$ are subspaces of V ,

$$0 \in W_i, i \in [1, k]$$

$$\therefore \text{Hence } 0_W = 0 + 0 + 0 + 0 + \dots$$

k times.

$$\therefore 0_W \in W$$

$$\therefore \vec{0} \in W$$

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11) Closure under +

for $x, y \in W$,

$$x = x_1 + \dots + x_k$$

$$y = y_1 + \dots + y_k$$

$$\therefore x+y = (x_1+y_1) + \dots + (x_k+y_k)$$

Now, for x_i and $y_i \in W_i$,

$$z_i = x_i + y_i \in W_i$$

$$\therefore x+y = z_1 + \dots + z_k \in (W_1 + W_2 + \dots + W_k)$$

$$\boxed{\therefore x+y \in W}$$

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11) Closure under scalar λ .

Let c be a scalar.

for $x \in W$,

$$x = x_1 + \dots + x_k$$

$$cx = c(x_1 + \dots + x_k)$$

$$cx = cx_1 + \dots + cx_k$$

Now, for $x_i \in W_i$
 $cx_i \in W_i$

$$\therefore cn \in (w_1 + \dots + w_n)$$

$$\boxed{cn \in W.}$$

$\therefore W$ is a subspace of V .

2). W is a subspace spanned by $w_1, v, \dots, v w_n$.

i). For $n \in W$

$$n = w_1 + \dots + w_k \text{ for}$$

$\therefore$ For $w_i \in W$,

$$\therefore w_i \in (w_1, w_2, \dots, v w_n)$$

$\therefore$ it belongs to one of those w_i .

$\therefore$ ~~$w_i \in$~~

$$w_1, w_2, \dots, w_k \in (w_1, v, \dots, v w_n)$$

$\therefore n$ is a linear combination (finite) of such w_i .

$$\therefore n \in \text{span}(w_1, v, \dots, v w_n).$$

9.09] As a vector in $\text{span}(w_1, v, \dots, v, w_k)$ belongs to a FINITE LINEAR COMBINATION of vectors ~~of~~ in this union.

So, for ~~each~~ ^{some} w_i in W_i ,

$$(w_1 + w_2 + \dots + w_k) \in \text{span}(w_1, w_2, \dots, w_k)$$

~~Now, for~~

$$\therefore \text{span}(w_1, w_2, \dots, w_k) \subseteq W.$$

$$\therefore \boxed{W = \text{span}(w_1, w_2, \dots, w_k)}$$

Hence proved.